YEAR 10 SCIENCE (EXTENSION)
PHYSICAL SCIENCE
TEST 2

Name: ______________________________ **Mark:** $\frac{}{64}$

Part 1: Multiple Choice

Write your choice for each question in the table below.

MULTIPLE CHOICE ANSWERS	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	

1. The crumple zone of a car is designed to:

   (a) allow the vehicle and passengers to stop more slowly.
   (b) decrease the time before the car comes to a stop after a collision.
   (c) allow the chassis to bend.
   (d) create a reaction force that equals the action force.

2. A car manufacturer wants to produce cars that accelerate faster. Which changes in their cars would result in greater acceleration?

   (a) Increase both the forward force by the engine and the mass of the car.
   (b) Decrease both the forward force and the mass of the car.
   (c) Increase the forward force and decrease the mass of the car.
   (d) Decrease the forward force and increase the mass of the car.

3. The rate of change of speed of an object is known as:

   (a) average speed.
   (b) average velocity.
   (c) change in velocity.
   (d) acceleration.

4. An object starts from rest and accelerates at 5.0 ms^{-2}. After 4.0 seconds, its velocity will be:

   (a) 0.
   (b) 10 ms^{-1}.
   (c) 20 ms^{-1}.
   (d) 30 ms^{-1}.

5. The tendency of an object to resist changes in its motion is called:

   (a) inertia.
   (b) weight.
   (c) friction.
   (d) work.

6. Newton's law, known as the law of inertia, is:

   (a) his first law.
   (b) his second law.
   (c) his third law.
   (d) none of the above.

7. An object is acted upon by a force of 50 N pushing it forwards and a total frictional force of 40 N acting in the opposite direction. The nett force will be:

(a) 10 N backwards.
(b) 10 N forwards.
(c) 90 N backwards.
(d) 90 N forwards.

8. The size of the acceleration acting on an object of mass 10 kg that has a nett force of 3.0 N acting on it will be:

(a) 13 ms^{-2}.
(b) 3.33 ms^{-2}.
(c) 0.3 ms^{-2}.
(d) 30 ms^{-2}.

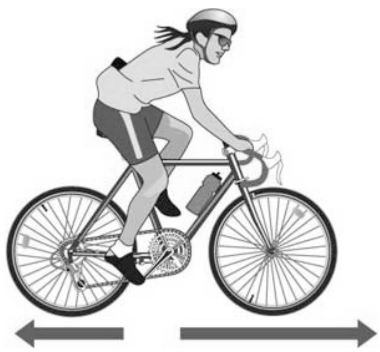
9. The force of gravity acting on an object is called:

(a) inertia.
(b) weight.
(c) friction.
(d) work.

10. The weight force acting on a 20.0 kg mass on the Earth would be:

(a) 19.6 N.
(b) 196 N.
(c) 1.96 N.
(d) 0 N.

11. In the diagram below, the arrows show the size and direction of two force arrows acting on the bike.



Which of the following describes the effect of the pair of forces on the bike?

(a) The bike will accelerate forward.
(b) The bike will change its direction of motion.
(c) The bike will remain stationary.
(d) The bike will decelerate.

12. What is the acceleration of a car if it increases its velocity from 5.0 ms^{-1} to 15.0 ms^{-1} in 4.0 seconds?

    (a) 2.5 ms^{-1}
    (b) 15 ms^{-2}
    (c) 2.5 ms^{-2}
    (d) 5.2 ms^{-2}

13. A class is studying Newton's three laws. The teacher asks a student wearing rollerblades to face the wall and push against it. The student moves backwards away from the wall. Which is the best explanation for her movement?

    (a) Her action of pushing resulted in a reaction force by the wall, in the opposite direction to her push.
    (b) She used her feet to push herself away from the wall.
    (c) The push is a force and causes her to accelerate.
    (d) The inertia of the student is responsible for her acceleration.

14. Which of the following statements is ***correct***?

    (a) A car at rest has at least two forces acting on it.
    (b) A car at rest has no forces acting on it.
    (c) When you take your foot off the car's accelerator, the car slows down because there are no more forces acting on it.
    (d) When you brake in a car that is rolling down a hill, the car stops because all of the forces add up to zero.

Part 2: Short Answer

Working to extended answer calculations should be set out clearly and logically.

1. (a) Which has the most inertia - a family car or a semi-trailer truck? Explain why. (2 marks)

   (b) Why is it not a good idea to pull in front of a such a truck as it is trying to stop at a set of traffic lights? (2 marks)

2. Describe the motion of a ball placed on the middle of the back seat of a car in the following situations.

   (a) The car brakes suddenly. ____ (1 mark)

   (b) The car turns right. ____ (1 mark)

   (c) The car accelerates. ____ (1 mark)

3. A dragster accelerates from rest to reach 325 kmh^{-1} after 6.75 s.

(a) What is the acceleration of the vehicle, assuming it is uniform? (3 marks)

(b) How far did the dragster cover in the first 5.00 s? (3 marks)

4. A 1400 kg Supercar enters the main straight of a race track at speed and accelerates at 8.35 ms^{-2} as it sprints for the finish line 300 m away, crossing it at 320 kmh^{-1}. What was the velocity of the car as it entered the straight? (3 marks)

5. A Space Shuttle lands at 270 km/h and immediately pops a drag chute to assist its brakes to stop the vehicle. Assume that the braking forces are uniform for this motion.

(a) If it takes 15.0 s to stop, calculate the ***deceleration*** experienced by the crew. (3 marks)

(b) If the mass of the Shuttle is 34,500 kg, what is the average force exerted by the brakes and the chute?
(If you could not find an answer for part (a), use deceleration = - 6.0 m/s^2.)
(3 marks)

(c) How far does the Shuttle travel while coming to a stop? (3 marks)

6. A car of mass 1750 kg is accelerated from 70.0 kmh^{-1} by the engine supplying a force of 2750 N. The frictional forces acting on the car total 375 N and the car travels in a straight line.

(a) What is the nett force acting on the car? (2 marks)

(b) How long does it take to reach 130 kmh^{-1}? (4 marks)

7. A 60.0 kg parachutist leaves a plane at 3000 m above the ground and free-falls to 1000 m before opening her parachute.

(a) What is her theoretical velocity when the parachute opens? (3 marks)

(b) This velocity is theoretical. Her velocity is actually 42.0 ms^{-1}. Why is this so different to your answer in part (a) ? (2 marks)

(c) The parachute decelerates the woman from 42.0 ms^{-1} to 17.0 ms^{-1} in 1.50 s. What force was applied by the parachute? (4 marks)

8. On the Moon, an object dropped from 1.00 m takes 1.23 s to reach the ground.

(a) What is the gravitational acceleration on the Moon? (3 marks)

(b) What is the weight of a 150 kg object on the Moon? **(If you didn't get an answer to part (a), use g = 1.69 ms^{-2}.)** (2 marks)

(c) Do you think you could lift the object off the ground if you were on the Moon? Explain your answer. (Assume you have a mobile space suit.) (2 marks)

.

9. You have researched a safety feature for this topic.

   (a) Name your safety feature.

   ____

   (b) Explain the main Physics principle behind how it acts to protect passengers.

   (3 marks)

YEAR 10 SCIENCE (EXTENSION)
PHYSICAL SCIENCES
DATA SHEET

Speed $\text{speed} = \frac{d}{t}$

Velocity $v = \frac{s}{t}$

Acceleration $a = \frac{v - u}{t}$

Equations of motion

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Force F = ma

Weight $F_w = mg$

Acceleration due to gravity g = 9.80 ms^{-2} (on Earth)

Converting kmh^{-1} to ms^{-1} Divide the speed in kmh^{-1} by 3.6 to get the speed in ms^{-1}.